

GTSC2143 Machine Learning for Business

Tutorial 9

Please write down your answers in this document and submit it at iSpace by the end of this tutorial.

Activity 1. Data Loading and Preprocessing

1. Load the Dataset

```
Load the BBC News dataset using pandas:

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.cluster import KMeans
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.decomposition import PCA
import seaborn as sns

# Load the BBC News Classification dataset
data = pd.read_csv("https://raw.githubusercontent.com/suraj-deshmukh/BBC-Dataset-News-Classification/refs/heads/master/dataset/dataset.csv", encoding='iso8859-1')
```

2. Explore the dataset:

- Display the first 3 rows
- Show the distribution of types
- Calculate average text length by category

Activity 2. Document Similarity and K-Nearest Neighbors

1. Create Document Representations

- Create TF-IDF representation using `TfidfVectorizer` (`max_features=1000`)
- Display the matrix shape

2. Measure Similarity

- Calculate cosine similarity between the first two news
- Find the 3 most similar documents to the first news in the dataset using `NearestNeighbors`
- Display their news text, types, and similarity scores

Activity 3. K-means Clustering

1. Apply Clustering

- Apply k-means with `k=5` clusters using TF-IDF matrix (`random_state=42`)
- Add cluster labels to the dataframe
- Show cluster distribution vs actual type distribution

2. Analyze the results

- For each cluster, display the top 10 most important words
- Based on top words, assign meaningful names to each cluster

- End of Tutorial 9 -